

Cytiva Xcellerex™ XDR-10 - 4.5-10 L working volume

cytiva-xdr-10

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	4.5 L	Source-supported
Maximum working volume	10 L	Source-supported
Nominal vessel / bag size	10 L	Source-supported
Total vessel volume loaded into the form	10 L	Derived / estimated
Initial working volume loaded into the form	4.5 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	1.5	App assumption
Internal / chamber diameter	0.204 m	Derived / estimated
Liquid height at maximum working volume	0.306 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	App assumption
Impeller count	1	App assumption
Impeller diameter	0.0673 m	Derived / estimated
Power number	0.4	App assumption
Initial agitation	100 rpm	App assumption
Maximum agitation	250 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 4.5 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 1; XDR family introduction

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 4.5 L.

Limitations: Explicit XDR-10 range. The separately archived perfusion application note is not an XDR-10 datasheet.

[Local original: Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 1](#)

[Manufacturer publication](#)

Maximum working volume - 10 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 1; XDR family introduction

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Explicit XDR-10 range. The separately archived perfusion application note is not an XDR-10 datasheet.

[Local original: Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 1](#)

[Manufacturer publication](#)

Nominal vessel / bag size - 10 L

Source-supported | Cytiva Xcellerex XDR 50 to 2000 L data file - PDF page 1; XDR family introduction

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Explicit XDR-10 range. The separately archived perfusion application note is not an XDR-10 datasheet.

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[Manufacturer publication](#)

Total vessel volume loaded into the form - 10 L

Derived / estimated | Form-population rule

Rule: total volume = max(nominal vessel volume, maximum working volume)

Selection / calculation: $\max(10 \text{ L}, 10 \text{ L}) = 10 \text{ L}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 4.5 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(4.5 \text{ L}, 10 \text{ L}) = 4.5 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 1.5

App assumption | Vessel-specific preset override

Rule: heightDiameterRatio = 1.5

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic mammalian stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.204 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V)/(\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.01 \text{ m}^3)/(\pi \times 1.5)} = 0.2039888 \text{ m}$; the form displays 0.204 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.306 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.2039888 \text{ m} \times 1.5 = 0.3059832 \text{ m}$; the form displays 0.306 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Pitched-blade turbine

App assumption | Vessel-specific preset override

Rule: impellerType = "pitched"

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic mammalian stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 1

App assumption | Vessel-specific preset override

Rule: impellerCount = 1

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic mammalian stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.0673 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.2039888 \text{ m}, 0.001 \text{ m}) = 0.0673163 \text{ m}$; the form displays 0.0673 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 0.4

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].powerNumber` = 0.4

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N_3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 100 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].baseRpm = 100

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 250 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].maxRpm = 250

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].baffled = false

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].maxPressure = 0.5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

The 1.5 H/D and single pitched-blade impeller are model-specific source-supported fields.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Cytiva Xcellerex XDR 50 to 2000 L data file

cytiva-xcellerex-xdr-50-to-2000-l-data-file-d7b9c0ca50bd.pdf

SHA-256: d7b9c0ca50bd6d6eff70df4dbd3d693d2c0a710badff26f8dece1f2333125f96